

Project Design Phase

Solution Architecture

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Name:	Varsaa Vihasini S B
Project Name:	Automated Network Request Management System
Maximum Marks:	2 Marks

Comprehensive Solution Architecture Narrative

❑ Streamlined System Flow (Left-Pane Architecture Narrative)

The operational runtime path follows a strict 5-stage automated trajectory engineered to maximize processing speed and remove human hand-off errors:

Ingestion & Load Phase: Corporate requesters open the custom Service Portal catalog view. The system instantly triggers lookups to auto-fill identity fields, ensuring data accuracy.

Sanitization Phase: The user inputs network metrics. Client-side validation blocks execution if it catches syntax formatting errors.

Trigger Phase: The form metadata reaches the server, prompting the backend workflow engine to process business logic paths.

Authorization Phase: Flow Designer spawns an interactive manager task card, holding the request in check until a signature event fires.

Commit Phase: Once authorization metrics are matched, validated properties execute persistent writes directly into our custom data storage directory.

❑ Level 1 Data Flow Diagram Mapping (Right-Pane Architecture Narrative)

The engineering layout utilizes a clean separation of roles between client-side interactions and core processing tables to keep the platform scalable and robust:

Requesters & Approvers (External Entities): Requesters interact solely with front-end form parameters. Authorized corporate managers process approval tasks from centralized workspace views without accessing backend tabs.
The Logic Core (Process 1.0): The primary automation hub reads user inputs, acts as a traffic controller, handles condition testing, and splits structural parent request request properties into distinct task units.

Relational Storage Layer (Data Stores): Baseline session info maps directly to system configuration directories. Executed engineering variables are securely isolated inside our custom ledger registry (u_network_database),

while unalterable compliance logs write directly to native platform audit streams (sys_audit).

External Integration Extensions: To support seamless operations, the backend incorporates the REST API Spoke framework. This provides secure external pathways out to third-party assets, inventory management databases, or orchestration layers for automated infrastructure changes.

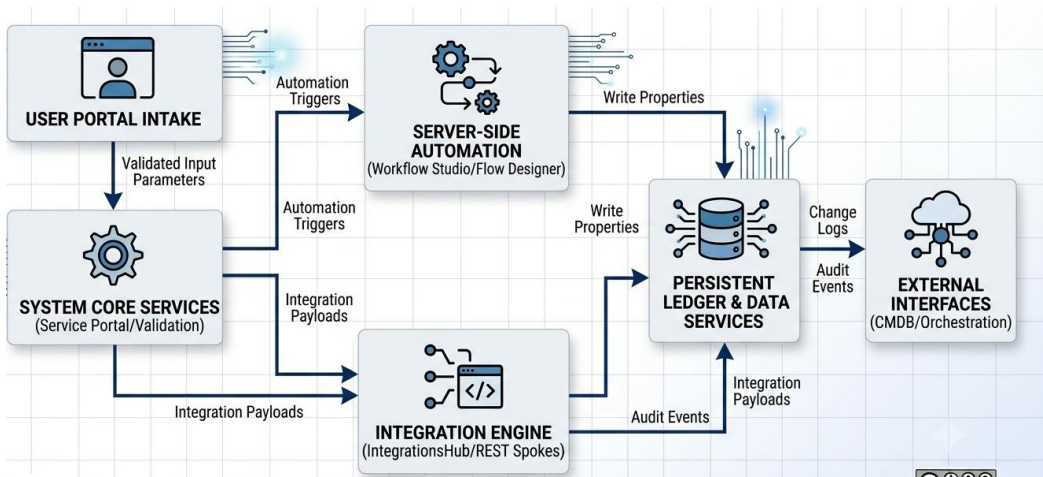


Table-1: Solution Architecture Structural Components

Use this formal table matrix to define the engineering boundaries, active processing jobs, and targeted database directories within your solution schema.

S.No	Process Module / Entity	Upstream Data Input	Core Execution Engine Tasks	Downstream Data Output
01	Service Portal Ingestion (User Interface)	Raw requester parameters (Form entries, login criteria).	Loads custom catalog structures, initializes active user session data blocks, and presents variables.	Sanitized variable strings passed to the validation client runtime.
02	Client-Side Validation (Front-End)	Form payload vectors from user input strings.	Runs real-time syntactic checks via AngularJS Client Scripts and active platform UI Policies.	Secured, error-free transaction arrays dispatched to backend layers.

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03	Flow Designer Routing <i>(Workflow Studio)</i>	Ingested transaction metadata parameters.	Evaluates business rules, queries hierarchical directory records, and routes interactive manager approval tasks.	Signed authorization event flags pushed to technical queue scripts.
04	Task Decomposition <i>(Engineering Workspace)</i>	Confirmed approval events and technical parameters.	Decomposes parent tickets into distinct child engineering task lines assigned to u_network_task.	Allocated task tickets sent to network group fulfillment queues.
05	Persistent Ledger Store <i>(Back-End Storage)</i>	Closed-complete task states and infrastructure metrics.	Executes secure write parameters, commits configurations to u_network_database, and logs sys_audit records.	Immutable historical assets and active portal dashboard updates.